Pocket FT8 Transceiver Revisited

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The Pocket FT8 Transceiver Revisited is a self-contained, portable software defined QRPP transceiver.

Introduction and Features

Charles Hill's original Pocket FT8 project[1] inspired several[2][3] fully self-contained FT8 transceiver derivatives in which the computer *is* the radio. This derivative revisits the original concept with simplified hardware and enhanced firmware retaining the pocket-sized, single-band, QRPP, HF transceiver concept for backpacking and day hike adventures. Features include:

- Open source hardware and firmware
- Self-contained and light-weight
- Compact, 4-layer PCB (4.0X2.8")
- FSK FT8 modulation
- 275+ mW RF output (QRPP)
- RF filter slot
- Low jitter TCXO
- GPS disciplined date, time and location (grid square)
- 480x320 pixel TFT calibrated touchscreen
- Waterfall traffic display
- FT8 QSO sequencer
- Automatic QSO logging to an ADIF file
- JSON station configuration file
- Operation from a single 5V USB power block

The FT8 protocol[4], designed for weak signal communication is ideal for portable, battery-operated, QRPP operation. While traditional setups consist of a computer, transceiver, and, in some cases, an audio interface, Pocket FT8 integrates these functions in a single, self-contained package. The use of FT8 reduces the transceiver's power requirements and weight, eliminating the telegraph key, microphone, and headset accessories for SSB/CW transceivers. Consolidating the radio with the FT8 encoder/decoder in an embedded computer eliminates the need to pack a computer in addition to the radio. Further integration of a GPS receiver provides the UTC date and time for ADIF logging as well as the Maidenhead grid square and FT8 time-synchronization. Of course, FT8 QSOs are terse, contest-like exchanges conveying sequenced, structured messages with minimal operator intervention after contact has been established. While FT8 is not always appropriate, commercial alternative designs[5][6] cover many of those situations.

Hardware Overview

The Pocket FT8 transceiver (Figure 1) is designed around a Teensy 4.1 600 mHz, 32-bit ARM Cortex-M7 MCU with hardware floating point, USB, SPI, I2C, 1 mB of RAM, and two ADCs.

When transmitting, the MCU reprograms the Si5351 clock generator's output for each of the eight FT8 Frequency Shift Keying (FSK) tones, a direct FSK scheme, unlike the Audio FSK (AFSK) widely used with a SSB transmitters' microphone jack. Frequency stability is achieved by locking the Si5351 to a low jitter Temperature Controlled Crystal Oscillator (TCXO) reference. The Si5351's output is buffered by one element of a SN74ACT244 octal bus driver whose remaining seven buffers function in parallel as the transmitter's QRPP power amplifier (PA). A five-pole Chebyshev low-pass filter (Figure 2) with 1 dB of ripple peaking near the expected operating frequency provides harmonic rejection. The PA delivers 275 mW through the filter.

The Si4735 implements the receiver's RF section and delivers FT8 audio tones to the MCU's audio pipeline via ADC2. While the Si4735 lacks the performance of alternative designs, its capabilities compliment the QRPP transmitter: on-the-air experience found the receiver decodes more signals than the transmitter can work, especially in crowded band conditions. The development of an early prototype discovered the Si4735 performance degrades in the presence of noise from other I2C bus device traffic; the problem was resolved by moving the Si4735 to a private I2C bus. The Si4735's phase-locked loop (PLL) is locked to one of the Si5351 clock generator's outputs.

An Adafruit 3.5" TFT 320x480 touchscreen with an HX8357D controller serve as the graphical user interface device. The display achieves satisfactory performance over an SPI bus from the MCU while the resistive touchscreen connects to the MCU's ADC1. While the compact display is ideal for portable operation, touchscreen accuracy improves with use of a stylus.

FL1 Design

FL1 mainly provides harmonic rejection for the transmitted signal. On 40 meters, the five pole Chebyshev design provides for 1 dB of ripple peaking inside the band.

During receive, the Chebyshev low-pass design offers only minimal rejection of out-of-band signals. While this has not proven to be a significant issue in the wilderness use-case, the receiver might benefit from a bandpass design in other applications.

Hardware Construction

T1

T1 is a broadband, bifilar wound transformer consisting of 10 turns of 26 AWG on an FT37-43 core.

FL1

The FL1 low-pass filter can be assembled and tested on a small daughter board. With some effort, it can probably be made pluggable to facilitate band changes.

Software

Software Design

Software Construction

Testing

Operation

Future

Acknowledgements

References

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[4] S. F. K. B. S. G. J. T. (K1JT), "The ft4 and ft8 communication protocols," *QEX*, July/August 2020.

[5] (2026, May). [Online]. Available: https://elecraft.com/products/kh1-transceiver

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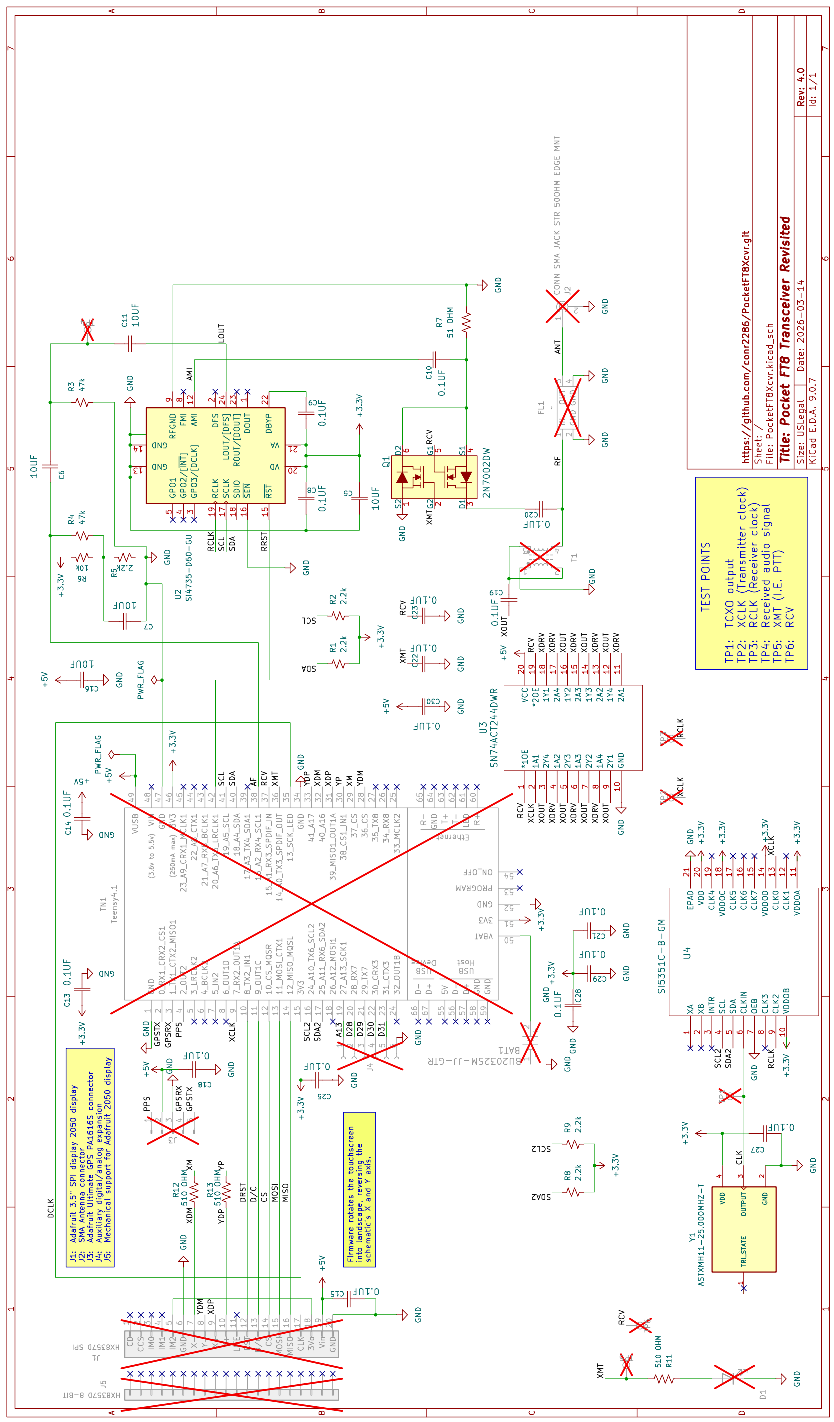

Figure 1: Schematic

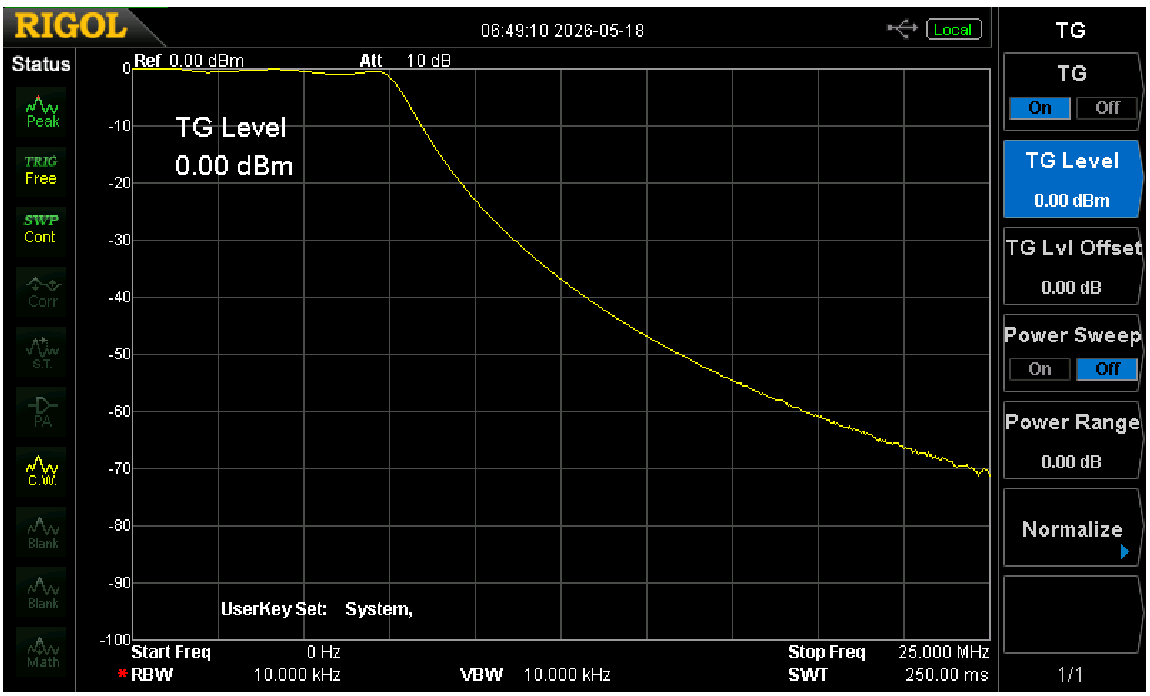

Figure 2: Chebyshev Low-Pass Filter Bode Plot